EXP_2026Q2_INVALID_SPLIT · SRM CHECK · ENGINEER

GROWTH · PRICING · 2026-06-05 · X² P = 0.0000

Sample ratio fails.

The observed split is too far from the expected split to trust the rest of the analysis. The verdict tree will fire INVALID-SRM at step 1; no metric value will be read.

> WHAT THIS GATE DOES
>
> **Halt. The analyzer does not proceed to metrics.**
>
> *A sample-ratio mismatch invalidates everything downstream — observed lift could just be a different population in each arm. The gate fires before any metric is read (R2).*

OBSERVED SPLIT

4229 / 5771

expected 5000 / 5000

X² STATISTIC	P-VALUE
237.78	0.0000
THRESHOLD	**VERDICT**
0.0005	BLOCK

READING THIS CHECK

A small p-value here invalidates the rest of the analysis.

When the χ^2 p-value is below the pre-registered threshold, the most likely explanation is an assignment-service bug, not a real treatment effect. The tree refuses to proceed to step 2 — the user must investigate the assignment service or override with a documented NoShipReasonCode.

COMPUTATION TRACE

srm_check is the only path to this number.

All SRM evaluations run through `agentxp.stats.srm.srm_check`; the function returns the trace below verbatim. Any change to the threshold or formula must come from a code change, not an ad-hoc call site (R4).

TEST	srm_chi_squared
OBSERVED	[4229, 5771]
EXPECTED	[5000, 5000]

CHI2_STAT	237.7764
P_VALUE	0.000000
THRESHOLD	0.0005
DF	1
VERDICT	BLOCK
FORMULA	chi^2 = sum((obs − exp)^2 / exp) · p from chi2(df=k−1)
